There Is A New World Order

With implants recording our experiences, and with the ability to play them back, will all experiences boil down to the right kind of implant?

Will we have any use for a "regular" life at all? Will we, as cyborgs, lie asleep, Matrix-style, experiencing the world via our implants? If thought control becomes possible, who will control whose thoughts? Who will be the ultimate controller, preventing us from bringing back slavery? If we can read each other's thoughts, we will not be able to lie-is that good or bad? Will the exorbitant prices of such implants mean that the rich will live forever and the poor will continue to die, relatively, young? Will there be exorbitantly-priced implants that offer control over other, lesser-priced variants?

These are profound questions, and they will become more and more important. A simple example: the Total Artificial Heart (*see box "Artificial Heart"*) costs $100,000 (Rs 44,61,000). Who gets to be in the queue?

The facts are simple: the technology exists; it will be improved. Immortality is just a few years or decades, and a few thousand successful experiments away—providing you can afford it!

What It Boils Down To...

So what does all this research amount to? What do Warwick's experiments, DeMarse's rat brain-controlled flight simulator, Karniel's lamprey controlled robot and the remote controlled cockroach have in common?

Apart from scientific curiosity, it is evolution. Human evolution has hit a plateau; we have reached the limit of interaction with the universe-you cannot move a cup just by thinking about it. Telekinesis may be beyond our reach, but biological cybernetics isn't. It is biological cybernetics that allows Warwick's computer to respond when he enters the room, or lets a disembodied brain, or parts of a brain, control a robot or a flight simulator.

As our brains develop, our bodies and the universe around us become bottlenecks. Soon we won't have to control hand muscles, and reach out for a mouse, move it to a Web link on our screen, and then control our finger muscles to click on it-instead, we will only "think it"! No longer will your life be dependant on how long your tissues and organs hold out; instead, you will get a brand new android body for your brain-your only concern will be visiting the "service centre" and "changing your oil"!

Ultimately, it boils down to controlling your own mortality, and achieving superhuman feats. We all want to live forever! ■

ram_mohan@thinkdigit.com
robert_smith@thinkdigit.com

Timeline

500 BC: Earliest known written reference to an artificial limb
300 BC: Capri, Italy: in 1858, a copper and wood leg dating from 300 BC unearthed
1863: Improvements to the attachment of artificial limbs
1898: Dr Vanghetti invents an artificial limb that moves through muscle contraction
1946: Major advancement in the attachment of lower limbs
1954: The first kidney transplant in Boston by Joseph Murray; also the first successful human organ transplant
1962: First breast Implants
1963: First liver transplant: the patient dies within a few days
1966: First pancreas transplant: immediately after the transplant, patient's blood sugar level begins to fall
1967: First Successful Liver Transplant. The liver functions for 13 months
1978: Rod Saunders implanted with the first research multichannel cochlear implant
1982: Graham Carrick implanted with the first commercially-available 22-channel Nucleus cochlear implant
1982: Dr Barney Clark implanted with the Jarvik-7, an artificial heart intended to last a lifetime. Patient survives 112 days
1988: First successful liver-bowel Transplant: a liver and six metres of intestine transplanted
1989: First combination heart, liver and kidney Transplant: Surgeons transplant a heart, liver, and kidney into a 26-year-old woman.
2000: "Smart" legs - entire smart lower limbs, with digital control systems - expected to be on the market in two years
2000: "Jerry," blind after a blow to his head 36 years prior, regains his ability to see thanks to an artificial eye. Jerry sees a simple display of dots that outline an object
2001: First completely self-contained artificial heart transplant
2003: The world's first brain prosthesis—an artificial hippocampus—is tested. Unlike devices such as cochlear implants, which merely simulate brain activity, this chip implant performs the same processes as the damaged part of the brain it is replacing